Aasharya Bhandari

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EDUCATION

B. Sc. Physics 2022- Present
Second year Percentage: 81.08 % (Top of the Class) St. Xavier's College, Tribhuvan University, Kathmandu, Nepal
Relevant Coursework: Classical Mechanics; Electrodynamics; Statistical physics ; Computational Physics; Probability and Inference.

RESEARCH EXPERIENCE

Kinetic Trajectory Simulation of Dusty Plasma
May 2025 – Present
Research Project, St. Xavier's College Maitighar, Kathmandu, Nepal
Supervisor: Dr. Roshan Chalise

- Computed modified Bohm sheath criterion for dusty plasma, then developed MATLAB code to include dusty particles incorporating Orbital Motion Limited (OML) theory.
- Studied variations in plasma density, particle behavior, and electric potential, comparing particle trajectory with and without the presence of dust particles.
- Showcased the importance of including dust effects in Kinetic Trajectory Simulations (KTS) method to accurately model its influence on plasma sheath dynamics.

Mathematical Modeling of Snow Mountain
Aug–Dec 2025
Intern Research Assistant, Phutung Research Institute, Tokha, Kathmandu, Nepal
Supervisor: Dr. Rijan Maharjan

- Worked on developing a mathematical model for the steady-state solution of snow accumulation and melting on mountainous terrain.
- Developed an Python-based solver to simulate the snow packed mountain dynamics.
- Contributed to parameterizing environmental factors such as precipitation rate, and solar radiation within the model framework.

SKILLS

Programming & Scripting: Python, C, C++, MATLAB, LATEX
Data Analysis : Pandas, ROOT, Pythia
Systems & Development: Linux, Git/GitHub

HONORS, AWARDS & GRANTS

Fathers Locke and Stiller Research Award (LSRA), St. Xavier's College, 2025
Undergraduate Merit Scholarship, St. Xavier's College, 2023–2025.
Academic Excellent Scholarship, New Horizon Institute, 2020-2022

COURSES, CONFERENCES, & WORKSHOPS

ICTP Physics without Frontiers Mini-Workshop on Particle Physics and Science Communication

Aug 9–11, 2023

Hands-on data analysis in High Energy Particle Physics. ICTP-PWF

Introduction to General Relativity

November 2024

Completed an introductory course on General Relativity, Tensor Calculus, and Differential Geometry St. Xavier's College

Introduction to Quantum Computing (2-Semester) 2023–2024

The Coding School, Qubit x Qubit

Introduction to Cirq, including quantum logic gates and essential quantum computing concepts and algorithms.

Completed a project on encryption and decryption using the BB84 Protocol.

PROJECTS AND PROGRAMMING EXPERIENCES

• Analyzed open ATLAS data using ROOT RDataFrame to plot the invariant Higgs mass from the 4-lepton decay channel.
• Simulated electron–positron collisions and calculated the invariant mass of the Z boson using Pythia and ROOT.
• Built a Deep Neural Network to classify Z boson decays and Higgs boson decays.
• Solved the double pendulum differential equation using the Runge–Kutta (RK4) method and visualized the results with matplotlib.
• Simulated proton–proton collisions using Pythia and analyzed ROOT files to create histograms for particle interaction studies.
• Developed a simulation of the ATLAS detector at CERN using Geant4 to model particle interactions and detector responses.